AP CALCULUS AB

Unit 1 • Lesson 1.6 practice • Answer key

Lesson 1.6 • Determining Limits Using Algebraic Manipulations

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to -2} \dfrac{x^2 - 4}{x + 2}$ is

(A) 0

(B) -4

(C) 4

(D) It does not exist.

Solution

Answer: (B). $\dfrac{(x-2)(x+2)}{x+2} = x - 2 \to -4.$

Question 2

NO CALCULATOR

$\lim_{x \to 4} \dfrac{\sqrt{x} - 2}{x - 4}$ is

(A) $\dfrac{1}{4}$

(B) $\dfrac{1}{2}$

(C) 0

(D) 4

Solution

Answer: (A). $x - 4 = (\sqrt{x} - 2)(\sqrt{x} + 2)$, so the quotient is $\dfrac{1}{\sqrt{x} + 2} \to \dfrac{1}{4}$.

Question 3

NO CALCULATOR

$\lim_{h \to 0} \dfrac{(3 + h)^2 - 9}{h}$ is

(A) 9

(B) 0

(C) 6

(D) 3

Solution

Answer: (C). $\dfrac{6h+h^2}{h} = 6+h \to 6.$

Question 4

NO CALCULATOR

$\lim\limits_{x\to 1} \dfrac{\frac{1}{x}-1}{x-1}$ is

(A) 1

(B) 0

(C) It does not exist.

(D) -1

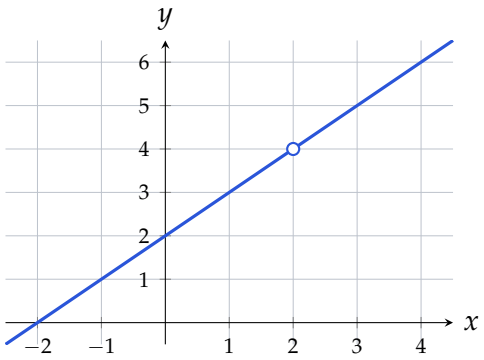
Solution

Answer: (D). $\dfrac{\frac{1-x}{x}}{x-1} = -\dfrac{1}{x} \to -1.$

Question 5 (graph)

NO CALCULATOR

The graph of $f(x) = \dfrac{x^2-4}{x-2}$ is shown. What is $\lim\limits_{x\to 2} f(x)$?



(A) 0

(B) 4

(C) 2

(D) It does not exist.

Solution

Answer: (B). The graph is the line $y = x+2$ with a hole at $(2,4)$. Algebraically, $\dfrac{(x-2)(x+2)}{x-2} = x+2 \to 4.$

Question 6 (Free response)

NO CALCULATOR

Evaluate each limit. Show the algebra.

(a) $\lim\limits_{x\to 5} \dfrac{x^2-3x-10}{x-5}$ (b) $\lim\limits_{x\to 0} \dfrac{\sqrt{4+x}-2}{x}$

(c) $\lim\limits_{x\to 2} \dfrac{x^3-8}{x-2}$ (d) $\lim\limits_{h\to 0} \dfrac{\frac{1}{2+h}-\frac{1}{2}}{h}$

Solution

(a) $\dfrac{(x-5)(x+2)}{x-5} = x+2 \to 7.$

(b) $\dfrac{(4+x)-4}{x(\sqrt{4+x}+2)} = \dfrac{1}{\sqrt{4+x}+2} \to \dfrac{1}{4}.$

(c) $x^3-8 = (x-2)(x^2+2x+4)$, so the limit is $4+4+4 = 12$.

(d) $\dfrac{\frac{2-(2+h)}{2(2+h)}}{h} = \dfrac{-1}{2(2+h)} \to -\dfrac{1}{4}.$